Second series question bank

1.What is a serial port terminal application? Mention any two applications and give examples.

2.Draw the block diagram of an I2C temperature sensor interfaced with an STM32 microcontroller and an LCD.

3.Differentiate between USART and I2C communication protocols.

4.Explain the purpose of USB HID class and mention the basic steps involved in creating a USB HID keyboard or mouse emulation device.

5. List the three MQTT Quality of Service (QoS) levels. Differentiate between MQTT and CoAP

6.Mention any three characteristics of LoRa communication. Why is LoRa suitable for IoT applications?

7.What is a queue in FreeRTOS? Mention one application.

8. Define memory optimization and state its importance in microcontrollers.

9.Explain how an STM32 microcontroller communicates with an SPI peripheral. Describe the sequence of operations involved in transmitting and receiving data.

10.With a neat block diagram, explain the configuration and implementation of CAN communication between multiple STM32U575 microcontrollers. Discuss CAN node configuration, CANH and CANL differential signaling, message identifiers, arbitration, error detection, CAN filters, and bus termination.

11. Explain the working principles of USART, I2C, SPI and CAN communication protocols. Compare them based on number of wires, communication method, speed and applications.

12. With a neat block diagram, explain the interfacing of an I2C temperature sensor and an LCD with an STM32U575 microcontroller. Describe the steps involved in sensor initialization, temperature-data acquisition, conversion, and LCD display.

13. Draw the layered architecture of IoT. Discuss the functions of sensing, network, processing, and application layers with suitable examples.

14.With a neat block diagram, explain ARM TrustZone architecture and its implementation in the STM32U575 microcontroller. Discuss Secure and Non-Secure worlds, memory and peripheral protection, and the configuration and management of Secure and Non-Secure regions.

15. Describe the steps involved in creating, scheduling and managing tasks in FreeRTOS on an STM32 microcontroller. Explain the purpose of task priorities and task states.

16. Explain the interfacing of a GSM module with an STM32 microcontroller. Describe the steps involved in making a call, sending an SMS and accessing internet services using suitable AT commands.